SEMINAR SUMMARY: APRIL 21, 2026 - BRENT CODY CONNECTIONS BETWEEN GRAPH THEORY AND MUSIC

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We start with an introduction to modular synthesizers. But we are quickly confronted with a problem about maximal evenness, presenting a 12-cycle C_{12}, we are curious about the maximal placing of 5 onsets over the cycle of 12. We try to space the onsets alternatively at first, and we get a section of three uncovered vertices in the cycle. We do a little fiddling, and we end up with the repeating pattern of the black-keys on a piano over the white keys on the piano. The set $J^0_{12,5} = \{0, 2, 4, 7, 9\}$ is the maximally even subset of C_{12}.

So naturally we ask, what does it mean to be maximally even? We come up with the following definition

Definition 1. (Clough, Douthett, 1991) Let $d(u, v)$ represent the distance from vertex u to vertex v of shortest length where $u, v \in C_n$, with C_n being the cycle on n-vertices. We call a set of vertices $\{a_0, a_1, \ldots, a_{m-1}\} \subseteq C_n$ *maximally even* if whenever there is $1 \leq k \leq \lfloor m/2 \rfloor$, we have

$$d(a_i, a_{i+k}) \in \left\{ \left\lfloor \frac{nk}{m} \right\rfloor, \left\lceil \frac{nk}{m} \right\rceil \right\}$$

Now we notice something curious, namely that

$$J_{12,5} = \left\{ \left\lfloor \frac{12i}{5} \right\rfloor : 0 \leq i < 5 \right\}.$$

So this theorem arises:

Theorem 1. *(Clough, Douthett, 1991) A set $A \subseteq C_n$ with $|A| = m$ is maximally even if and only if it is of the form*

$$A = J^r_{n,m} = \left\{ \left\lfloor \frac{ni + r}{m} \right\rfloor : 0 \leq i < m \right\}.$$

And it seems that quite naturally the complement of a maximally even set is also maximally even

$$C_n \setminus J^r_{m,n} = J^{n-r-1}_{n,m-n}$$

Through the reset of the talk we talked about minimal energy from maximal energy, a curiosity indeed. I suppose a system run at full throttle can only maintain itself for so long

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before buckling. After buckling a system can continue to run, but needs help and fixing after buckling has occurred. Think of an engine perhaps you throw a rod, well there are 5 cylinders still running, but are you going to get the same performance. By no means. The same it seems is true of humans.